

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Bioscience and Biotechnology
2. **Subject Name:** Biochemical Reaction Engineering and Bioenergetics
L-T-P: 3-1-0 **Credits:** 4
3. **Pre-requisites:** Biochemistry (BT11002)
4. **Syllabus and reference books:**

Syllabus:

1. Kinetics of homogeneous reactions, Equilibrium and Reaction rates; Models of enzyme catalysis
2. Kinetics of Enzyme-catalyzed reactions in free state, Michaelis-Menten Kinetics; Equilibrium and quasi steady-state assumptions
3. Inhibition of enzyme-catalyzed reactions, Effect of heat, pH, and toxins on enzyme stability
4. Kinetics of multi-substrate multi-product reactions; Sequential and Ping-pong reactions
5. Immobilized Enzyme kinetics; Effect of mass transfer; Boundary layer mass transfer; Intraparticle mass transfer
6. Kinetics of substrate utilization, product formation and biomass production
7. Monod growth model and its various modifications, Structured and unstructured kinetic rate Models
8. Death kinetics of cells and spores
9. Basic concepts of thermodynamics and Introduction to Bioenergetics
10. Gibbs free energy, chemical potential and biochemical reactions
11. Biological membranes, transport and membrane potential, Nernst equation
12. Energetics of ATP hydrolysis and biochemical roles
13. ATP synthesis, glycolysis, TCA cycle, electron transport, ATP synthase
14. Macromolecular interactions, free energy linking, coupling
15. Energy balance equations, enthalpy calculations, energy balance calculations, Thermodynamics of microbial growth, Energy balance equation for cell culture, Fermentation energy-balance examples

Reference Books:

- 1) Biochemical Engineering Fundamentals, by James E. Bailey and David F. Ollis, McGraw-Hill International, 2nd Edition.
- 2) Enzymes: A Practical Introduction to Structure, Mechanism, and Data Analysis, by Robert A. Copeland, John Wiley and Sons Inc., 2nd Edition
- 3) Principles of physical biochemistry, by K.E. van Holde, Curtis Johnson, Pui Shing Ho Prentice Hall, 2nd Edition

4) Voet's Principles of Biochemistry, by Donald Voet, Judith G. Voet, Charlotte W. Pratt
John Wiley & Sons; 5th Edition, ISBN-10: 1119451663

5) Bioprocess Engineering Principles, by Pauline M. Doran, Academic Press, 2nd
Edition

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Kinetics of homogeneous reactions, Equilibrium and Reaction rates; Models of enzyme catalysis	2
2.	Kinetics of Enzyme-catalyzed reactions in free state, Michaelis-Menten Kinetics; Equilibrium and quasi steady-state assumptions	3
3.	Inhibition of enzyme-catalyzed reactions, Effect of heat, pH, and toxins on enzyme stability	4
4.	Kinetics of multi-substrate multi-product reactions; Sequential and Ping-pong reactions	4
5.	Immobilized Enzyme kinetics; Effect of mass transfer; Boundary layer mass transfer; Intraparticle mass transfer	4
6.	Kinetics of substrate utilization, product formation and biomass production	2
7.	Monod growth model and its various modifications	2
8.	Structured and unstructured kinetic rate Models	2
9.	Death kinetics of cells and spores	2
10.	Basic concepts of thermodynamics and Introduction to Bioenergetics	2
11.	Gibbs free energy, chemical potential and biochemical reactions	3
12.	Biological membranes, transport and membrane potential, Nernst equation	5
13.	Energetics of ATP hydrolysis and biochemical roles	3
14.	ATP synthesis, glycolysis, TCA cycle, electron transport, ATP synthase	4
15.	Macromolecular interactions, free energy linking, coupling	4
16.	Energy balance equations, enthalpy calculations, energy balance calculations, Thermodynamics of microbial growth, Energy balance equation for cell culture, Fermentation energy-balance examples	4

Total number of hours	50
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